

ReDataX

M5: An Alternative Approach to Markout Estimation

Egor Vasilev

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1 Model role and testable hypothesis

Once negative markout values are clipped at zero, the target acquires a large point mass at $S = 0$, while its positive part remains continuous, asymmetric, and heavy-tailed to the right. M4 asks one regression to model both the frequency of zeros and the magnitude of rare large events. M5 separates these tasks in the spirit of hurdle models [1].

For target market A , time t , and horizon $H = 600$ seconds, define aggressor-aligned markout $m_{A,H}(t)$ and its positive part $S_{A,H}(t) = \max\{m_{A,H}(t), 0\}$. The model estimates three quantities:

$$\begin{aligned} p_A(t) &= \mathbb{P}(S_{A,H}(t) > 0 \mid X_A(t)), \\ \mu_A(t) &= \mathbb{E}[S_{A,H}(t) \mid S_{A,H}(t) > 0, X_A(t)], \\ q_A(t) &= \mathbb{P}(m_{A,H}(t) > m_{\text{BE}} \mid X_A(t)). \end{aligned}$$

The first two form the forecast $\hat{S}^{\text{M5}} = \hat{p} \cdot \hat{\mu}$; the third acts as an admissibility filter.

Hypothesis 1.1. Modeling the probability and conditional magnitude of positive markout separately produces a more useful expected-exposure score than direct M4 regression and, after costs and budget are taken into account, leads to a more valuable policy.

More concretely, the hypothesis says that P3 has positive absolute value relative to no action under P0, and that P3 outperforms both P2, the policy based on M4, and P1, the probability-ranking policy.

2 Why use a two-part model?

Regression on a zero-clipped label mixes two distinct signals: whether the event occurs and how large it is. If the classifier and regressor are trained on different subsets, each can be tuned to its own task. A classifier estimates whether markout will be positive, or whether it will clear the break-even threshold. The regressor estimates how large the move will be among positive cases. The policy then combines the estimates, giving it more room to manage the trade-off between intervention frequency and average payoff.

3 Data and target definitions

The implementation uses public Binance Spot `aggTrades`, aggregated on a one-second UTC grid for BTCUSDT, ETHUSDT, and ETHBTC. For the BTCUSDT and ETHUSDT targets, the direction of the hypothetical exposure is determined by the sign of current normalized aggressive flow, $d_A(t) = \text{sign } \phi_A(t)$.

Definition 3.1. The *aggressor-aligned markout* is

$$m_{A,H}(t) = 10^4 d_A(t) \frac{P_A(t+H) - P_A(t)}{P_A(t)},$$

where P_A is one-second VWAP. A positive markout means that price moved in the aggressor's direction, creating a potential loss for the hypothetical passive side.

Definition 3.2. The M5 labels are

$$Z_i = \mathbf{1}\{m_i > 0\}, \quad B_i = \mathbf{1}\{m_i > m_{\text{BE}}\}.$$

Z_i is the label for the positive-markout classifier; B_i is the label for a separate classifier that predicts whether markout exceeds the break-even threshold m_{BE} .

The internalization share is $\iota = 0.25$, protection efficiency is $\eta = 0.50$, and the action cost is $c = 0.50$ basis points. The protectable share is therefore $\rho = \iota\eta = 0.125$, and the break-even threshold is

$$m_{\text{BE}} = \frac{c}{\rho} = 4.0 \text{ bps.}$$

The primary horizon is 600 seconds and the decision stride is 10 seconds. An observation is admissible when at least 600 seconds have elapsed since the start of the day; the target market is active; $d_A(t) \neq 0$; notional is positive; VWAP is finite and positive at t and $t + H$; and $t + H$ remains within the same day.

4 Feature space

M5 uses the same 208-column feature matrix as the final M4 specification.

5 Model architecture

5.1 Probability classifiers

The probability of positive markout, $\hat{p}_i = \hat{\mathbb{P}}(Z_i = 1 \mid X_i)$, is estimated with a `HistGradientBoostingClassifier` using logistic loss. A second classifier with the same architecture estimates $\hat{q}_i = \hat{\mathbb{P}}(B_i = 1 \mid X_i)$. Under P3, the latter acts as an admissibility filter.

5.2 Conditional-magnitude regressor

The $\hat{\mu}$ regressor is trained only on observations for which $Z_i = 1$. Its target is log-transformed and capped:

$$y_i^+ = \log(1 + \min\{m_i, C\}), \quad C = 100 \text{ bps.}$$

After training, the prediction is transformed back:

$$\hat{\mu}_i = \text{clip}\{\exp(\hat{y}_i^+) - 1, 0, C\}.$$

The regression component uses `HistGradientBoostingRegressor` with squared-error loss. Observations are weighted by notional:

$$w_i = \text{clip}\left[\left(\frac{N_i}{\text{median}(N_{\text{train}})}\right)^{0.25}, 0.25, 4\right].$$

Definition 5.1. The final forecast of expected positive-markout magnitude is

$$\hat{S}_i^{\text{M5}} = \hat{p}_i \cdot \hat{\mu}_i.$$

This decomposition follows from the law of total expectation. In the current version, the gradient-boosting probabilities are not separately calibrated by isotonic or logistic calibration, and $\hat{\mu}_i$ is not corrected for inverse-log retransformation bias [6]. The product $\hat{p} \cdot \hat{\mu}$ should therefore be read as a practically useful expected-exposure score.

5.3 Hyperparameters

The selected `medium` preset uses a learning rate of 0.05, 160 iterations, up to 31 leaf nodes per tree, at least 60 observations per leaf, an L_2 -regularization coefficient of 1, 63 histogram bins, and a target cap of $C = 100$ bps. Random states are fixed and early stopping is disabled. The gradient-boosting methods are described in [5, 7].

6 Final configuration selection

The 600-second horizon is the most stable during development and remains positive in validation for both markets.

Table 1: M5 selection path: mean daily value, USDT per 1m USDT.

Market	Split	Mean	R	Positive days	Status
BTCUSDT	development, 600 s	15.77	13.29	100%	selected
BTCUSDT	validation, 600 s	16.80	9.15	100%	selected
ETHUSDT	development, 600 s	28.70	18.69	100%	selected
ETHUSDT	validation, 600 s	30.33	20.60	100%	selected

Both markets use the `medium` preset, a 10% budget, and a safety margin of $\delta = 0.10$ bps. The probability threshold τ is 40% for BTCUSDT and 0 for ETHUSDT.

7 Final results

Table 2: P3 policy over the final period, 10–16 February 2025.

Metric	BTCUSDT	ETHUSDT
Pooled net value, USDT per 1m USDT	12.86	24.76
Mean daily value, USDT per 1m USDT	9.25	20.08
95% interval for mean difference vs. P0	[2.85; 18.05]	[11.52; 32.31]
Daily standard deviation	10.62	14.77
Worst day, USDT per 1m USDT	-1.88	5.10
Share of positive days	85.7%	100.0%
Notional touched	9.93%	10.00%
Exposure captured	17.21%	16.20%
Benefit-cost ratio	3.59	5.95
Share of P4 oracle value	26.6%	28.6%

Pooled value weights days by notional, whereas mean daily value assigns equal weight to every day. Both quantities are positive and statistically distinguishable from zero relative to no action.

Table 3: Paired daily P3 comparisons during the final week.

Market	Comparison	Mean difference	95% interval	Conclusion
BTCUSDT	P3 – P2 (M4)	4.54	[1.79; 7.06]	positive
BTCUSDT	P3 – P1	1.56	[-0.30; 3.44]	inconclusive
ETHUSDT	P3 – P2 (M4)	3.50	[-0.13; 7.47]	inconclusive
ETHUSDT	P3 – P1	-2.06	[-4.21; 0.24]	inconclusive

Result 7.1. P3 has positive value relative to no action in both markets. M5 outperforms direct M4 regression for BTCUSDT, but that advantage is not established for ETHUSDT. Superiority over the P1 probability policy is not established for either market.

8 Limitations and reproducibility

M5 inherits the limitations of the preceding models. The parameters ι , η , and c come from the literature and are not estimates for a particular financial institution. Changing any of them changes the break-even threshold and, therefore, the B_i filter. The `aggTrades` data describe public Binance flow but contain no order book, client toxicity, or actual execution and hedging costs. Neither $\hat{p}$ nor $\hat{q}$ receives separate probability calibration, and the inverse-log transformation of $\hat{\mu}$ is not bias-corrected. The final period consists of seven calendar days, which limits generalization across market regimes and rare tail events. M5 does not dominate every alternative policy universally; the choice of model still depends on whether the goal is to maximize absolute protected value or efficiency per unit of intervention.

References

- [1] J. G. Cragg. Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica*, 39(5), 829–844, 1971.
- [2] L. R. Glosten, P. R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100, 1985.
- [3] D. Easley, M. M. López de Prado, M. O’Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493, 2012.
- [4] R. Cont, A. Kukanov, S. Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1), 47–88, 2014.
- [5] J. H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232, 2001.
- [6] N. Duan. Smearing estimate: a nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383), 605–610, 1983.
- [7] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830, 2011.

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- [8] Binance. Spot REST API: Compressed/Aggregate Trades List. <https://developers.binance.com/en/docs/catalog/core-trading-spot-trading/api/rest-api/market>.
 - [9] B. Efron, R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.